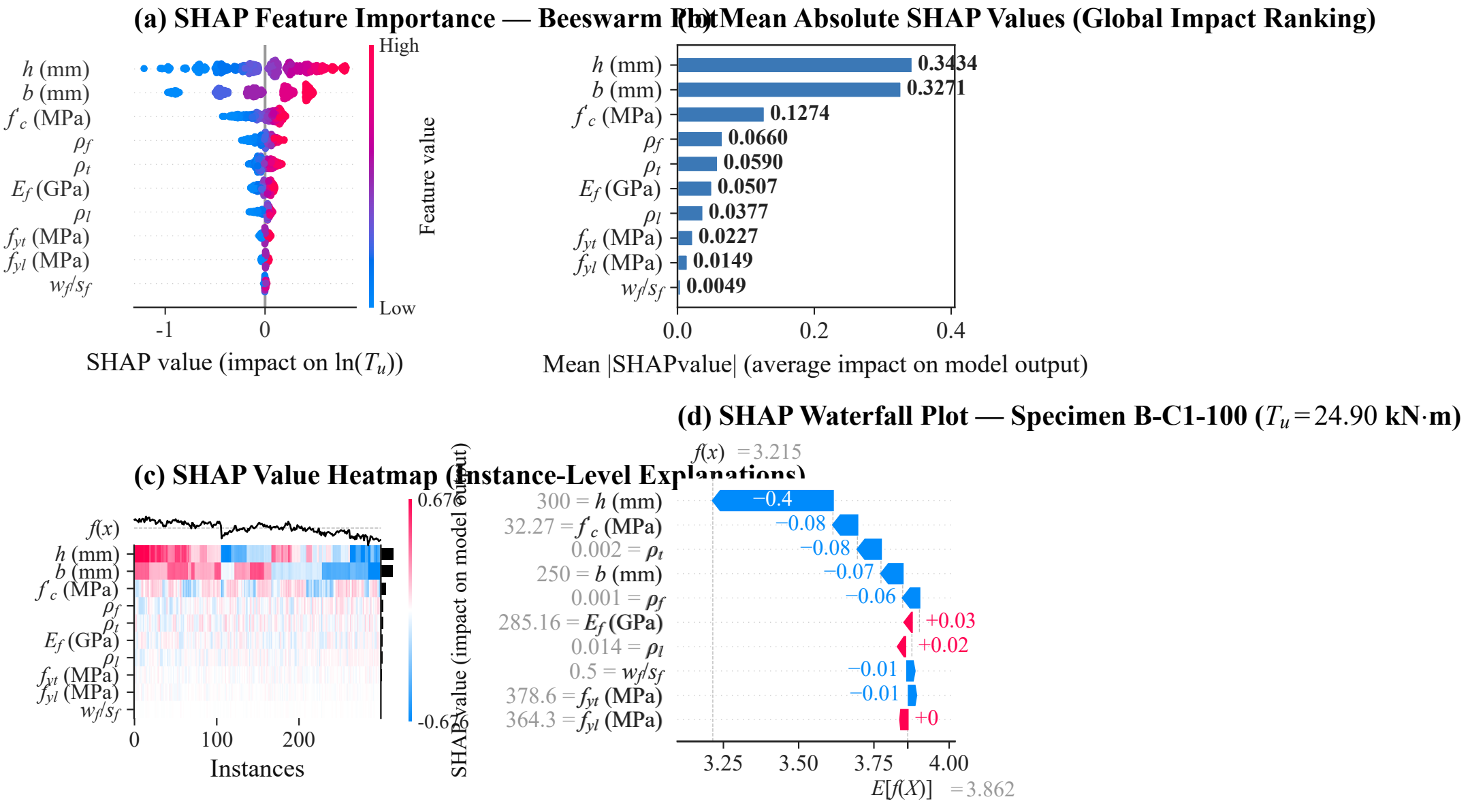

(a) SHAP Feature Importance — Beeswarm Plot
h (mm)
b (mm)
f′c (MPa)
ρf
ρt
Ef (GPa)
ρl
fyt (MPa)
fyl (MPa)
wf/sf
High
Low
Feature value
-1
0
SHAP value (impact on ln(Tu))
(b) Mean Absolute SHAP Values (Global Impact Ranking)
0.3434
0.3271
0.1274
0.0660
0.0590
0.0507
0.0377
0.0227
0.0149
0.0049
0.0
0.2
0.4
Mean |SHAPvalue| (average impact on model output)
(c) SHAP Value Heatmap (Instance-Level Explanations)
f(x)
0.676
-0.676
0
100
200
Instances
SHAP value (impact on model output)
(d) SHAP Waterfall Plot — Specimen B-C1-100 (Tu = 24.90 kN·m)
f(x) = 3.215
300 = h (mm)
32.27 = f′c (MPa)
0.002 = ρt
250 = b (mm)
0.001 = ρf
285.16 = Ef (GPa)
0.014 = ρl
0.5 = wf/sf
378.6 = fyt (MPa)
364.3 = fyl (MPa)
−0.4
−0.08
−0.08
−0.07
−0.06
+0.03
+0.02
−0.01
−0.01
+0
3.25
3.50
3.75
4.00
E[f(X)] = 3.862